A/B Testing & Hypothesis Testing Report

Workora Labs Data Analytics Internship

Dataset: checkout_ab_test_v2.csv. The analysis evaluates checkout conversion performance between Control (A) and Variant (B).

1. Experiment Summary

Metric	Control (A)	Variant (B)
Visitors	600	600
Conversions	57	71
Conversion rate	9.50%	11.83%

2. Hypotheses

H0 (Null): There is no difference in conversion rates between Control A and Variant B.

H1 (Alternative): The conversion rates of Control A and Variant B are different.

Significance level (alpha): 0.05 (5%).

3. Conversion Analysis

Control A converted 57 of 600 visitors (9.50%). Variant B converted 71 of 600 visitors (11.83%). The observed absolute difference is 2.33 percentage points, equivalent to a relative lift of 24.56% for B compared with A.

4. Two-Proportion Z-Test

Statistic	Result
Z-statistic	1.3092
Two-sided p-value	0.1905
Alpha	0.05
95% CI for B – A	-1.16 pp to 5.82 pp

Interpretation: the p-value is greater than 0.05, so the observed difference does not meet the selected 5% threshold for statistical significance. The confidence interval also includes zero, so the data do not establish a statistically reliable difference between the variants.

5. Chi-Square Test Cross-Check

The 2×2 Variant × Converted table was also tested using a Chi-Square test of independence. Chi-square = 1.4780, degrees of freedom = 1, p-value = 0.2241. This cross-check leads to the same conclusion at the 5% significance level.

6. Effect Size

Absolute conversion-rate difference (B – A): 2.33 percentage points.
Relative conversion lift: 24.56%.
95% confidence interval: -1.16 to 5.82 percentage points.

7. Executive Decision Memo

Decision: Keep Control A for the current experiment.

Variant B shows a higher observed conversion rate than Control A (11.83% vs 9.50%), but the difference is not statistically significant at the 5% level (p = 0.1905). Therefore, this experiment does not provide sufficient statistical evidence to conclude that Variant B produces a reliable conversion improvement. A follow-up experiment with a larger sample or a longer test period may be considered if additional evidence is needed.

8. Final Result

Item	Result
Control conversion	9.50%
Variant conversion	11.83%
Observed relative lift	24.56%
Z-test p-value	0.1905
Statistically significant at 5%?	No
Recommended experimental status	Keep Control A

Note: This report uses the supplied synthetic Workora Labs dataset. Statistical conclusions apply to this experiment and its measured sample, not to a broader population without further validation.